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Inventor(s)

Haberman; Seth et al.

SYSTEMS, METHODS AND COMPUTER-READABLE MEDIA FOR RESOURCE-BASED ALLOCATION OF CONTENT TRANSMITTED IN A MEDIA NETWORK

Abstract

Systems, methods and computer-readable storage media for allocating content elements addressed for a content opportunity in a transport stream within a household-addressable media network. A content opportunity may include an advertising slot in a programming stream. The content elements may include advertising content configured to be run during the advertising slot. The advertising slot may be transmitted in a programming stream providing television programming and the advertising content may be provided in a commercial feed. The media network may be configured to transmit various content formats, such as high definition (HD), standard definition (SD), and variations thereof. Systems may be configured to allocate the highest profile corresponding content elements if there is sufficient resources available. If sufficient resources are not available, the system may allocate lower profile versions of the content elements.

Inventors: Haberman; Seth (New York, NY), Niemeijer; Gerrit (Maplewood, NJ)

Applicant: FreeWheel Media, Inc. (Philadelphia, PA)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/503,106, filed Nov. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/712,949, filed Apr. 4, 2022, now U.S. Pat. No. 11,849,161, which issued on Dec. 19, 2023, which is a continuation of U.S. patent application Ser. No. 16/037,966, filed Jul. 17, 2018, now U.S. Pat. No. 11,323,759, which issued May 3, 2022, which is a continuation of U.S. patent application Ser. No. 14/866,364, filed Sep. 25, 2015, now U.S. Pat. No. 10,057,610, which issued on Aug. 21, 2018, which is a continuation of U.S. patent application Ser. No. 13/961,238, filed Aug. 7, 2013, now U.S. Pat. No. 9,148,683, which issued on Sep. 29, 2015, which claims the benefit of U.S. Provisional Application No. 61/680,450 filed on Aug. 7, 2012, each of which is incorporated herein by reference in its entirety.

FIELD OF INVENTION

[0002] The described technology generally relates to allocation of bandwidth in transport streams broadcast within a media network, and, more specifically, to allocating of bandwidth for delivery of content having different resource profiles based on the available resources of the transport streams.

BACKGROUND

[0003] Media networks may simultaneously broadcast multiple types of programming streams, such as high definition (HD) and standard definition (SD) programming streams. In general, HD programming may have a 16:9 aspect ratio and a resolution of about 1920 vertical lines of resolution by 1080 horizontal lines of resolution, or 1920×1080 (2.1 megapixels per frame) or 1280×720 (0.9 megapixels per frame). Standard definition (SD) is generally used to refer to signal formats that are lower than HD. Media networks may provide various types of content in the available formats based on user demand, content provider request, network requirements. The different formats require different resources, for example, HD content requires more bandwidth than SD bandwidth. In general, media networks increase resource utilization in order to provide higher profile content, such as transmitting HD content using additional bandwidth. However, increasing resource utilization to accommodate content introduces inefficiencies and decreases the profitability of providing multiple types of programming formats. Accordingly, media networks would benefit from methods configured to efficiently allocate bandwidth for content for transmission to viewers using existing resources.

SUMMARY

[0004] This disclosure is not limited to the particular systems, devices and methods described, as these may vary. The terminology used in the description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.

[0005] As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. Nothing in this disclosure is to be construed as an admission that the embodiments described in this disclosure are not entitled to antedate such disclosure by virtue of prior invention. As used in this document, the term “comprising” means “including, but not limited to.”

[0006] Embodiments of the present invention provide for adaptive bandwidth management to allocate bandwidth resources between content of various formats and resolutions to ensure delivery of all content assets to the receiving end of a transmission. Embodiments of the invention determine the required bandwidth for delivery of scheduled or targeted content, and if insufficient resources, such as bandwidth, are unavailable, manage existing resources and content such that all content is assured of delivery through available content streams or channels.

[0007] In an embodiment, a system for allocating of bandwidth for delivery of content elements addressed for a content opportunity in a transport stream within a household-addressable media network may include a processor and a non-transitory, computer-readable storage medium in operable communication with the processor. The computer-readable storage medium contains one or more programming instructions that, when executed, cause the processor to detect a content opportunity in a transport stream, determine a plurality of content elements addressed for the content opportunity, allocate of bandwidth for a highest available profile version of each of the plurality of content elements to a highest profile available content stream responsive to the highest profile available content stream having sufficient resources, and allocate bandwidth for a lower profile version of each of the plurality of content elements remaining after allocation of the bandwidth for the highest profile available version of each of the plurality of content elements to a lower profile content stream.

[0008] In an embodiment, a computer-implemented method for allocating bandwidth for content elements addressed for a content opportunity in a transport stream within a household-addressable media network may include, by a processor, detecting a content opportunity in a transport stream, determining a plurality of content elements addressed for the content opportunity, allocating bandwidth for a highest available profile version of each of the plurality of content elements to a highest profile available content stream responsive to the highest profile available content stream having sufficient resources, and allocating bandwidth for a lower profile version of each of the plurality of content elements remaining after allocation of bandwidth for the highest profile available version of each of the plurality of content elements to a lower profile content stream.

[0009] In an embodiment, a computer-readable storage medium having computer-readable program code configured to allocate bandwidth for content elements addressed for a content opportunity in a transport stream within a household-addressable media network, the computer-readable program code may include computer-readable program code configured to detect a content opportunity in a transport stream, computer-readable program code configured to determine a plurality of content elements addressed for the content opportunity, computer-readable program code configured to allocate bandwidth for a highest available profile version of each of the plurality of content elements to a highest profile available content stream responsive to the highest profile available content stream having sufficient resources, and computer-readable program code configured to allocate bandwidth for a lower profile version of each of the plurality of content elements remaining after allocation of bandwidth for the highest profile available version of each of the plurality of content elements to a lower profile content stream.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other objects of the present invention will become more readily apparent from the following detailed description taken in connection with the accompanying drawings.

[0011] FIG. 1 depicts an illustrative media network according to some embodiments.

[0012] FIG. 2 depicts a flow diagram for an illustrative bandwidth allocation method configured according to some embodiments.

[0013] FIGS. 3A-3C depict illustrative bandwidth allocation systems according to some embodiments.

[0014] FIG. 4 depicts a block diagram of exemplary internal hardware that may be used to contain or implement the various computer processes and systems as discussed above.

DETAILED DESCRIPTION

[0015] The described technology is generally directed to systems, methods and computer-readable media for allocation of resources for content addressed for a content opportunity in a broadcast stream. A media network may be configured to transmit content to customer premises equipment (CPE) in multiple formats or resolutions having different resource requirements, such as a high profile format and a low profile format. For instance, the high profile format may require more bandwidth within a transport stream than the low profile format. A non-limiting example of a high profile format is high definition (HD) content, while a non-limiting example of a low profile format is standard definition content (SD). Additional formats or resolutions may include, but are not limited to, 4K, 8K, 1080i, 1080p, 780i, 780p, or the like. According to some embodiments, the media network may be configured to provide high profile content in response to requests for high profile content until a resource threshold has been met. After the resource threshold has been reached, the media network may provide requested content in a low profile format. In this manner, the media network may provide the requested content without requiring additional resources, such as additional bandwidth.

[0016] FIG. 1 depicts an illustrative media network according to some embodiments. As shown in FIG. 1, a media network **100** may include a media content provider **105**. Non-limiting examples of a media content provider **105** include a television broadcast network, a cable television network, a satellite television network, an internet service provider (ISP), a computing device advertising network, a media distribution network, a cloud computing network, a local area network (LAN), a wide area network (WAN), terrestrial network, mobile network, 1-way network, 2-way network, or any combination thereof. The media content provider **105** may transmit content to one or more local content systems **110a-110n** configured to communicate with an audience **115** of the media network **100**. The local content systems **110a-110n** may include equipment and systems configured to transmit media content received from the media content provider **105** to a defined portion of the audience **115**. Illustrative and non-restrictive examples of a local content system **110a-110n** include a cable television network headend, an internet service provider base station, or the like.

[0017] According to embodiments, the media content provider **105** may be configured to operate across physical device platforms and networks simultaneously. For example, content may be delivered to CPEs, such as set-top-boxes (STBs), Smart televisions (SmartTV) and/or digital video recorders (DVRs), over a cable television system, to mobile computing devices using standard network communication protocols (for instance, Ethernet or Wi-Fi) over an ISP network, to smart phone devices over standard telecommunication protocols (for instance, third Generation (3G), fourth Generation (4G), long-term evolution (LTE), or the like), and to home gateway devices through a LAN, WAN and/or ISP network.

[0018] The audience **115** may include a plurality of households **120** capable of receiving media content from the media network **100** through various receiver devices or customer premises

equipment (CPE), including, without limitation, a STB, a television, a DVR, a personal computer (PC), a laptop computer, a mobile computing device, a smartphone, a tablet computing device, a home gateway, or the like. Although multiple households **120** are depicted in FIG. **1**, only one is labeled to simplify the figure. The audience **115** may be divided into segments **125a-125n** based on various segmentation factors. Non-limiting examples of segmentation factors include age, gender, occupation, years of home ownership, household size, income, geographic location, family size, media consumption habits, and combinations thereof. For example, an audience segment may include males between the ages of 25 to 35 with an annual income above \$60,000.

[0019] According to some embodiments, the audience may be household addressable such that the media content provider **105** and/or the local content systems **110a-110n** may direct particular content directly to each household **120** and/or segments **125a-125n** of the audience **115**. According to some embodiments, content may generally include any type of data capable of being received and consumed by a recipient. Illustrative and non-restrictive examples of content include advertising, entertainment programs, informational programs, messages, video, audio, graphical, and/or animated content. The media content provider **105** may direct particular media content, such as an advertisement or a television program, only to certain segments **125a-125n**.

[0020] In an embodiment, the media content provider **105** and/or the local content systems **110a-110n** may be configured to implement linear addressable systems, such as a linear addressable television advertising system. In a linear addressable system, content may be targeted to individual household **120** receiver devices, such as STBs, home gateways, and/or DVRs in a linear broadcast system. In general, a linear broadcast system may be configured to broadcast all or substantially all content (for example, programming networks) to all receiver devices within the audience **115**.

[0021] Providing targeted content or targeted content elements, such as advertisements, within a linear broadcast system may require that receiver devices are able to access the targeted content during the time that the targeted content is scheduled to be shown, which may generally be referred to as an “avail.” For example, targeted content elements may be transmitted on separate bandwidth or transport streams, such as a commercial feed stream.

[0022] The media content provider **105** may be configured to simultaneously broadcast program content to CPEs in multiple formats or profiles, including, without limitation, standard definition (SD) and high definition (HD) formats. The program content may be broadcast using a programming stream, such as an MPEG-2 encoded transmission stream, configured for the particular format of the content.

[0023] The programming stream may include content opportunities (placement opportunities, “avails,” commercial slots, advertising slots, slots) for switching the programming stream being transmitted to the CPE to a different stream (a “content stream”) to receive a content element. The CPE may switch back to the programming stream responsive to the completion of the content element. For example, a television programming stream may have a content opportunity which is configured as a “commercial break” in the programming. During the content opportunity, the CPE may switch from receiving the television programming stream to receiving a content stream, such as a commercial feeder stream configured to broadcast commercial content. After the content elements (for example, television commercials) in the commercial stream have completed, the CPE may switch back to the television programming stream. In general, the media network is configured to provide content elements having a profile that corresponds with the profile associated with the content opportunity. For example, a HD commercial may be presented for an avail on a HD programming stream.

[0024] In a household-addressable media network **100**, multiple content elements may be transmitted for each content opportunity. For example, three content elements addressed to three different segments **125a-125n** may be broadcast using one or more content streams. The appropriate content element may be selected for viewing by the user, for example, at the local content system **110a-110n** level and/or at the CPE. As such, the media content provider **105** may

broadcast multiple programming transport streams and multiple content streams for various types of formats. For example, a particular television network may be broadcast within the media network **100** over one HD programming stream and one SD programming stream. In addition, the media network **100** may transmit at least one HD content stream and one SD content, for instance, for commercials targeted for avails in the HD and SD programming streams.

[0025] The transmission of programming and content elements using multiple transport streams may have an effect on the resource requirements and efficiency of the media network **100**. For example, adding resources, such as bandwidth through the addition of transport streams, may require equipment, network assets, and resource utilization requirements that may impact the cost-effectiveness of the media network **100** and the level of service provided to users. As such, media content providers **105** are limited in their ability to increase bandwidth in response to increased demand.

[0026] Providing household-addressable content, such as television advertisements, in high profile formats according to conventional technology is generally implemented by increasing the available bandwidth, for instance, by transmitting content using additional transport streams. For example, a media network may be configured to provide HD and SD programming. A HD programming stream may be configured to transmit two television networks. The HD programming stream may include a content offering that the media content provider has sold to an advertiser requiring three household-addressable HD advertisements. A corresponding HD content stream for transmitting the HD advertisements may similarly only transmit two commercials during the same content offering slot. According to existing technology, the media network may either fail to transmit one of the three HD advertisements or transmit the three HD advertisements using two HD content streams (the first two HD advertisements on one HD content stream and the third HD advertisement on the second HD content stream). However, such systems are not flexible and are not optimized for resource utilization, such as the allocation and use of network bandwidth. For instance, once the HD avail starts, a system configured according to conventional technology may either increase resource utilization or partially fail the household-addressable avail.

[0027] Accordingly, some embodiments described herein may be configured to dynamically adapt the allocation and/or the format of bandwidth for content elements addressed for a content offering. FIG. 2 depicts a flow diagram for an illustrative bandwidth allocation method configured according to some embodiments. In various embodiments, the components described herein with respect to FIG. 2 may generally be combined in any order and/or any combination, are not limited by the order described herein, and may contain more or fewer components. The steps described in FIG. 2 are described as being carried out by a system, which may include various components of a media network, such components of media network **100** depicted in FIG. 1, including one or more computing devices, servers, network equipment components, software applications, or the like.

[0028] As shown in FIG. 2, a system may detect **205** a content opportunity in a transport stream. For example, the content opportunity may be an avail for a television program in a programming stream. The programming stream may include various formats, such as a high profile format, including an HD format. In general, the format of the programming stream will not be the lowest format available in the media network. For instance, the media format may be an HD transport stream in a media network configured to transmit HD and SD transport streams. The system may determine **210** the content elements addressed for the content opportunity. For example, multiple household-addressable commercials may be targeted for the content slot, such as three versions of a television commercial aimed at three different media network audience segments.

[0029] The system may allocate bandwidth **215** for the content elements in available content streams. For example, the media network may include one high profile (for example, HD) content stream and one low profile (for example, SD) content stream for transmitting content elements. In an embodiment, the content streams may be configured as commercial feeds within the media network. The system may determine **220** whether there are resources available in the highest

profile available content stream. For example, resources may include bandwidth, number of channels, bitrate, or other limiting aspect of a transport stream. In general, the highest profile available content stream may be the content stream having a profile that corresponds with the profile of the programming stream associated with the content opportunity. For example, the content opportunity may be in a high profile programming stream, such as an HD programming stream. As such, the highest profile available content stream may include a high profile (HD) programming stream.

[0030] If the system determines **220** that resources are available, then bandwidth for a highest profile available content element is allocated **225** in the highest profile available content stream; otherwise, bandwidth for a lower available content element is allocated **230** in a lower profile available content stream. For instance, if the transport stream is a HD programming stream, then the system may determine **220** whether enough resources are available in a HD commercial stream. If there are sufficient resources, the system may allocate **225** bandwidth for an HD commercial in the HD commercial stream at a time corresponding with the content opportunity. If there are not sufficient resources, the system may allocate **230** bandwidth for a SD commercial in a SD commercial stream at a time corresponding with the content opportunity.

[0031] In this manner, a media network system may have the flexibility to allocate bandwidth for content elements within existing bandwidth and/or limit the expansion of bandwidth by allocating bandwidth for content having various profiles. As described in reference to FIG. 2, some embodiments provide for a media network system that may dynamically allocate bandwidth for content having a different profile than the original profile associated with the requested content in order to manage system resources while maintaining the delivery of requested content.

[0032] In an embodiment, the system may generate and/or have access to content elements of available profiles and may select therefrom when allocating **225**, **230** bandwidth for content elements. In another embodiment, the system may dynamically change, modify, transcode, or otherwise transform the profile of a content element in order to allocate bandwidth for the content element in a content stream. For example, the system may be configured to encode a HD content element as a SD content element as required. According to some embodiments, the system may be configured to modify the profile of multiple content elements in order to meet certain resource requirements. For instance, if three SD content elements are encoded at about 4 Mbps and only 10 Mbps of bandwidth is available, the system may transcode all three SD content elements to about 3.33 Mbps.

[0033] The system may determine **235** whether bandwidth for all content elements have been allocated. If the system determines **235** that all bandwidth for content elements have been allocated, the allocation process is complete **240** for the content opportunity; otherwise, the system may continue to allocate **215** bandwidth for the content elements into the content streams.

[0034] In an embodiment, a media network may allocate bandwidth for content elements based on one or more factors in addition to the methods described according to embodiments described herein. For example, embodiments may be configured to prioritize content elements, content opportunities, segments, or the like so that bandwidth for certain content elements have a higher priority for being allocated in high profile versions as compared to lower priority content elements. According to various embodiments, content elements may be prioritized based on price, segment demographics, content type, or the like. For example, a media network may sell content opportunities to content providers at a higher rate in order to have bandwidth for content elements targeted therefor have a higher priority for being allocated as a high profiler version. In another example, the media network may prioritize bandwidth for certain segments for being allocated higher profile versions, such as a segment known to have a HD television may be prioritized over a segment less likely to have a HD television or viewers in an age group of 25-45 being prioritized over an age group of 65 and above. In a further example, content elements may be prioritized based on content type, for instance, program promotions may be prioritized over product advertisements.

According to such embodiments, the system may determine which content element to allocate first so that it may have the highest probability of being allocated in a high profile version.

[0035] Although HD and SD profiles have been used as examples herein, embodiments are not so limited, as any type of media profile, format, system, compression, encoding level, or the like, that may be used according to some embodiments is contemplated herein. For example, a media network may include multiple profiles based on encoding levels, such as HD at 6 Mbps, HD at 9 Mbps, HD at 11 Mbps, and so on.

[0036] In one example, a media network may use three formats, a first format, a second format, and a third format, each with various resource requirements. In this example, the first format requires the most of a particular resource, the third format requires the least of a particular resource, and the second format requires the resource at a level between the first and the third. According to some embodiments, a media network may attempt to allocate bandwidth for content elements having a certain profile (for instance, the profile associated with the programming stream that includes the content opportunity) and, if sufficient resources (for instance, bandwidth) are not available, the system may attempt to allocate bandwidth for content elements having a lower profile.

[0037] In this example, the media network may attempt to allocate bandwidth for four content elements in a first profile content stream in response to detecting a content opportunity in a first profile programming stream. The first profile content stream may have sufficient resources for the first content element. As such, the media network may allocate bandwidth for a first profile version of the first content element in the first profile content stream. The first profile content stream may not have sufficient resources for the second content element and the media network may attempt to allocate bandwidth for the second content element in a second profile content stream. The second profile content stream may have sufficient resources for the second content element. The media network may then allocate bandwidth for a second profile version of the second content element in the second profile content stream. The first profile content stream and the second profile content stream may not have sufficient resources for the third content element. As such, the media network may attempt to allocate bandwidth for the third content element in the third profile content stream. The third profile content stream may have sufficient resources for the third content element.

Accordingly, the media network may allocate bandwidth for a first profile version of the third content element in the third profile content stream. The media network may determine that the first profile content stream now has sufficient resources for the fourth content element and may allocate bandwidth for a first profile version of the fourth content element in the first profile content stream.

[0038] FIGS. 3A-3C depict an illustrative content allocation system according to some embodiments. As shown in FIG. 3A, a SD programming stream **305a** and a HD programming stream **310a** may be transmitted within a media network, such as media network **100** depicted in FIG. 1. According to some embodiments, the SD programming stream **305a** and the HD programming stream **310a** may broadcast SD and HD versions, respectively, of the same television network. Avails **315a**, **315b** may be transmitted over the HD programming stream **305b** and the SD programming stream **310a**. The avails **315a**, **315b** may be slots for the same commercial purchased by the same advertiser. As depicted in FIG. 3A, the avail **315a** for the HD programming stream **305a** may be delayed compared to the avail **315b** for the HD programming stream **305b** due to certain characteristics of the HD format that require longer processing and/or transmission times as compared with SD programming.

[0039] In general, SD content is encoded at a lower bitrate as compared with HD content, as such, the HD programming stream **305b** may have fewer networks per transport stream as compared with the SD programming stream **305a**. In an embodiment, the SD programming stream **305a** and the HD programming stream **310a** may include MPEG-2 transport streams. In the example of FIG. 3A, the HD content is encoded 2:1 (for example, 2 HD streams in 1 transport stream) and the SD content is encoded at 5:1.

[0040] In the example embodiment depicted in FIG. 3A, three household-addressable

advertisements have been targeted for the avails **315a**, **315b**. For example, two different types of a first commercial addressed to a first and a second audience segment, respectively, and a third commercial addressed to a third segment. The SD household-addressable advertisements **325a-325c** associated with the SD avail **315a** may be allocated to a SD commercial feed **310a**. An HD commercial feed **310b** only has two streams for allocating content. As such, bandwidth for HD versions of commercials **320a** and **320b** are allocated to the HD commercial feed **310b**. The media network allocates bandwidth for a SD version of commercial **320c** to an available stream in the SD commercial feed.

[0041] In an alternative embodiment, as depicted in FIG. 3B, the bandwidth for avail **315c** is allocated to the SD version of commercial **320c** contained within the HD programming transport stream. As described above, this may occur when the number of targeted HD avails exceeds the HD bandwidth and one or more of the avails needs to be stepped down to a SD format. Similarly, as depicted in FIG. 3C, the bandwidth of avails **315a-315c** is allocated, in the manner described above, to SD versions of commercials **325a-325c** and inserted to the HD programming stream.

[0042] According to one embodiment of the invention, SD and HD commercials may be inserted in real-time into the programming streams according to a pre-defined schedule, such as scheduled commercial blocks, set off by cue tones or other triggers, to be displayed on a device tuned to the programming stream. Alternatively, one embodiment provides for determining available bandwidth in the programming transport streams to deliver content for playback at a later time. Such content may be buffered or stored for a longer time in a data store at the receiving end of the transmission, such as the STB, television memory, mobile device memory, or the like. The content stored in such a manner may include instructions for the local device to assemble, insert or playback the content at a later time.

[0043] FIG. 4 depicts a block diagram of exemplary internal hardware that may be used to contain or implement the various computer processes and systems as discussed above. A bus **400** serves as the main information highway interconnecting the other illustrated components of the hardware. CPU **405** is the central processing unit of the system, performing calculations and logic operations required to execute a program. CPU **405**, alone or in conjunction with one or more of the other elements disclosed in FIG. 4, is an exemplary processing device, computing device or processor as such terms are used within this disclosure. Read only memory (ROM) **430** and random access memory (RAM) **435** constitute exemplary memory devices.

[0044] A controller **420** interfaces with one or more optional memory devices **425** to the system bus **400**. These memory devices **425** may include, for example, an external or internal DVD drive, a CD ROM drive, a hard drive, flash memory, a USB drive or the like. As indicated previously, these various drives and controllers are optional devices. Additionally, the memory devices **425** may be configured to include individual files for storing any software modules or instructions, auxiliary data, common files for storing groups of results or auxiliary, or one or more databases for storing the result information, auxiliary data, and related information as discussed above.

[0045] Program instructions, software or interactive modules for performing any of the functional steps associated with, among other things, analyzing a transport stream, inserting targeted content into a transport stream, and storing targeted data as described above may be stored in the ROM **430** and/or the RAM **435**. Optionally, the program instructions may be stored on a tangible computer-readable medium such as a compact disk, a digital disk, flash memory, a memory card, a USB drive, an optical disc storage medium, such as a Blu-ray™ disc, and/or other recording medium.

[0046] An optional display interface **430** may permit information from the bus **400** to be displayed on the display **435** in audio, visual, graphic or alphanumeric format. The information may include information related to a current job ticket and associated tasks. Communication with external devices may occur using various communication ports **440**. An exemplary communication port **440** may be attached to a communications network, such as the Internet or a local area network.

[0047] The hardware may also include an interface **445** which allows for receipt of data from input

devices such as a keyboard 450 or other input device 455 such as a mouse, a joystick, a touch screen, a remote control, a pointing device, a video input device and/or an audio input device. [0048] It will be appreciated that various of the above-disclosed and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. It will also be appreciated that various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art which alternatives, variations and improvements are also intended to be encompassed by the following claims.

Claims

1. A method comprising: receiving, by a user device of a plurality of user devices, a linear content stream; detecting, by the user device, an advertisement placement opportunity in the linear content stream; and outputting, by the user device, an advertisement having video resolution based on resource information associated with the linear content stream, wherein the resource information facilitates output by each user device of the plurality of user devices.
2. The method of claim 1, wherein the resource information comprises at least one of: information associated with bandwidth of the linear content stream, information associated with limiting aspects of the linear content stream, or information associated with bitrate of the linear content stream.
3. The method of claim 1, wherein a second user device, of the plurality of user devices, output the advertisement having a second video resolution, different from the video resolution, based on the resource information.
4. The method of claim 1, wherein the plurality of user devices comprise one or more of: set-top boxes; Smart televisions; or smartphones.
5. The method of claim 1, wherein the plurality of user devices further comprise a plurality of different user device platforms comprising: a set-top box; and a Smart television.
6. The method of claim 1, wherein the linear content is delivered to the plurality of user devices via a plurality of different network platform types comprising: a cable television network; and an Internet Service Provider network.
7. The method of claim 1, wherein the linear content is received via Internet Protocol.
8. The method of claim 1, wherein the advertisement comprises an addressed advertisement addressed to each user device of the plurality of user devices.
9. A non-transitory computer-readable medium storing instructions that, when executed, cause: receiving, by a user device of a plurality of user devices, a linear content stream; detecting, by the user device, an advertisement placement opportunity in the linear content stream; and outputting, by the user device, an advertisement having video resolution based on resource information associated with the linear content stream, wherein the resource information facilitates output by each user device of the plurality of user devices.
10. The non-transitory computer-readable medium of claim 9, wherein the resource information comprises at least one of: information associated with bandwidth of the linear content stream, information associated with limiting aspects of the linear content stream, or information associated with bitrate of the linear content stream.
11. The non-transitory computer-readable medium of claim 9, wherein a second user device, of the plurality of user devices, output the advertisement having a second video resolution, different from the video resolution, based on the resource information.
12. The non-transitory computer-readable medium of claim 9, wherein the plurality of user devices comprise one or more of: set-top boxes; Smart televisions; or smartphones.
13. The non-transitory computer-readable medium of claim 9, wherein the plurality of user devices further comprise a plurality of different user device platforms comprising: a set-top box; and a

Smart television.

14. The non-transitory computer-readable medium of claim 9, wherein the linear content is delivered to the plurality of user devices via a plurality of different network platform types comprising: a cable television network; and an Internet Service Provider network.

15. The non-transitory computer-readable medium of claim 9, wherein the linear content is received via Internet Protocol.

16. The non-transitory computer-readable medium of claim 9, wherein the advertisement comprises an addressed advertisement addressed to each user device of the plurality of user devices.

17. A system comprising: a first user device, of a plurality of user devices, configured to: receive a linear content stream; detect an advertisement placement opportunity in the linear content stream; and output an advertisement having video resolution based on resource information associated with the linear content stream, wherein the resource information facilitates output by each user device of the plurality of user devices; and a second user device, of the plurality of user devices, configured to: output the advertisement having a second video resolution, different from the video resolution, based on the resource information.

18. The system of claim 17, wherein the resource information comprises at least one of: information associated with bandwidth of the linear content stream, information associated with limiting aspects of the linear content stream, or information associated with bitrate of the linear content stream.

19. The system of claim 17, wherein the plurality of user devices comprise one or more of: set-top boxes; Smart televisions; or smartphones.

20. The system of claim 17, wherein the plurality of user devices further comprise a plurality of different user device platforms comprising: a set-top box; and a Smart television.

21. The system of claim 17, wherein the linear content is delivered to the plurality of user devices via a plurality of different network platform types comprising: a cable television network; and an Internet Service Provider network.

22. The system of claim 17, wherein the linear content is received via Internet Protocol.

23. The system of claim 17, wherein the advertisement comprises an addressed advertisement addressed to each user device of the plurality of user devices.
